

CSA0921 - Programming in Java

Exception Handling Programs

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1.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=845`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Vidhushini T S" with their registration number "192511061".

The main content area is titled "EXCEPTION-06/08/26". On the left, a "QUESTIONS" sidebar lists four questions, all marked as answered (4/4 answered):

- Q1 Handling ArithmeticException (10 marks)
- Q2 Handling ArrayIndexOutOfBoundsException (10 marks)
- Q3 Handling NumberFormatException (10 marks)
- Q4 Handling Multiple Exceptions (10 marks)

The selected question, "Q1 Handling ArithmeticException", is displayed in the main area. It asks: "Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message." (10 marks).

The "Your Code" section shows the following Java code:

```
1 import java.util.Scanner;
2 public class Divide{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             int a=sc.nextInt();
7             int b=sc.nextInt();
8             int c=a/b;
9             System.out.println("Result: "+c);
10        }catch(ArithmeticException e){
11            System.out.println("Error: Division by zero not allowed");
12        }
13    }
14 }
```

The "Input" section shows the user entered "10" and "0". The "Output" section shows the result: "Error: Division by zero not allowed".

At the bottom of the interface, there are "Run" and "Save" buttons. The Windows taskbar at the very bottom shows the system clock as 14:34 on 06-08-2026, with a temperature of 35°C and "Partly sunny" weather.

2.

SIMATS Online Programming Lab
JAVA PROGRAMMING

EXCEPTION-06/08/26

QUESTIONS

- Q1 Handling ArithmeticException 10 marks
- Q2 Handling ArrayIndexOutOfBoundsException 10 marks
- Q3 Handling NumberFormatException 10 marks
- Q4 Handling Multiple Exceptions 10 marks

4/4 answered

Q2 Handling ArrayIndexOutOfBoundsException 10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2 public class Array{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int[] a={10,20,30,40,50};
6         try{
7             int index=sc.nextInt();
8             System.out.println("Element: "+a[index]);
9         }catch(ArrayIndexOutOfBoundsException e){
10             System.out.println("Error: Index out of bounds");
11         }
12     }
13 }
```

Input

8

Output

Error: Index out of bounds

Run Save

3.

SIMATS Online Programming Lab
JAVA PROGRAMMING

EXCEPTION-06/08/26

QUESTIONS

- Q1 Handling ArithmeticException 10 marks
- Q2 Handling ArrayIndexOutOfBoundsException 10 marks
- Q3 Handling NumberFormatException 10 marks
- Q4 Handling Multiple Exceptions 10 marks

4/4 answered

Q3 Handling NumberFormatException 10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;
2 public class Converst{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             String str=sc.next();
7             int num=Integer.parseInt(str);
8             System.out.println("Number: "+num);
9         }catch(NumberFormatException e){
10             System.out.println("Error: Invalid number format");
11         }
12     }
13 }
```

Input

abc

Output

Error: Invalid number format

Run Save

4.

SIMATS Online Programming Lab
JAVA PROGRAMMING

Vidhushini T S
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Q1

Handling ArithmeticException

10 marks

Q2

Handling ArrayIndexOutOfBoundsException

10 marks

Q3

Handling NumberFormatException

10 marks

Q4

Handling Multiple Exceptions

10 marks

4/4 answered

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```

1 import java.util.Scanner;
2 public class MultipleBlock{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int[] arr=new int[2];
6         try{
7             int a=sc.nextInt();
8             int b=sc.nextInt();
9             int index=sc.nextInt();
10            arr[index]=a/b;
11            System.out.println("Result: "+arr[index]);
12        }catch(ArithmeticException e){
13            System.out.println("Error: Division by zero not allowed");
14        }catch(ArrayIndexOutOfBoundsException e){
15            System.out.println("Error: Index out of bounds");
16        }
17    }
18 }

```

Input

10 0 0

Output

Error: Division by zero not allowed

Run

Save

35°C

Partly sunny

Search

ENG

IN

14:35

06-08-2026